

$$C_0 = \langle a, b, c \rangle \cong G \times G \times G$$

(coef. of [a], coef of [b], coef of [c])

$$C_1 = \langle [ab], [bc], [ac] \rangle \cong G^3$$

$$C_2 = \emptyset \,, \quad C_p = \emptyset \ \forall\, p \geq 2$$

$$\partial_0 : C_0 \longrightarrow C_{-1} := \{\emptyset\}$$

$$x \longmapsto \emptyset$$

$$\ker(\partial_0) = C_0$$

$$\mathrm{Im}(\partial_0) = \{[\emptyset]\}$$

↑ one elt is $\emptyset$ + it is hiy.

$$\partial_2 : C_2 \longrightarrow C_1$$

↑ elt

$$\emptyset \longmapsto \phi = \text{☺}\cdot[ab] + \text{☺}[bc] + \text{☺}[ac]$$

where ☺ is the id in G

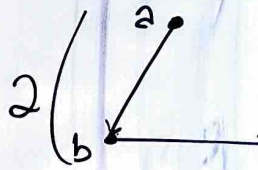
$$\ker(\partial_2) = \{\emptyset\} = C_2$$

$$\mathrm{Im}(\partial_2) = \{\emptyset\} \subseteq C_1$$

$\partial_1 : C_1$

[ab]
[bc]
[ac]

[ca
= −[ac



$\partial_1 : C_1 \to C_0$

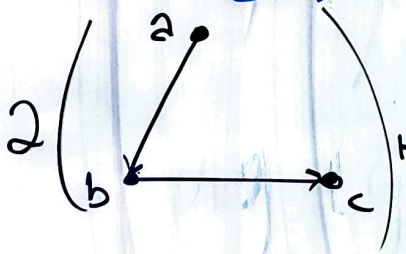
$[ab] \longmapsto b-a$

$[bc] \longmapsto c-b$

$[ac] \longmapsto c-a$

specify where my generators go (one oriented copy of each edge/simplex)

$[ca] = -[ac] \longmapsto \partial(-[ac]) = -\partial([ac]) = -(c-a) = a-c$



$\partial([ab]+[bc]) = \partial([ab]) + \partial([bc]) = b-a+c-b = c-a$

all of these can be figured out from where the generators mapped

$\cdot[ab]+\cdot[bc]+\cdot[ac]$ where $\cdot$ is the id in G

$$\text{im}(\partial_1) = \{x \in C_0 \mid \exists\, x' \in C_1 \text{ s.t. } \partial_1(x') = x\}$$

$$= \langle \partial([a,b]), \partial([b,c]), \partial([a,c]) \rangle$$

↑ sometimes we drop the subscript when obvious

$$= \langle [b]-[a], [c]-[b], [c]-[a] \rangle$$

$$= \langle [b]-[a], [c]-[b] \rangle \quad \text{b/c } [c]-[a] = ([b]-[a])+([c]-[b])$$

$$c-a = (b-a)+(c-b)$$

sometimes brackets get clunky

$$\cong G^2$$

$$\ker(\partial_1) = \{x \in C_1 \mid \partial_1(x) = \phi\}$$

$$= \{g[ab] + g[bc] + g[ca] \mid g \in G\}$$

$$= \langle [ab] + [bc] + [ca] \rangle \cong G$$

$$= \langle [ab] + [bc] - [ac] \rangle$$

$Z_2 = \ker(\partial_2) = C_2 = \{\emptyset\}$ trivial group

$B_1 = \text{Im}(\partial_2) = \{\emptyset\}$

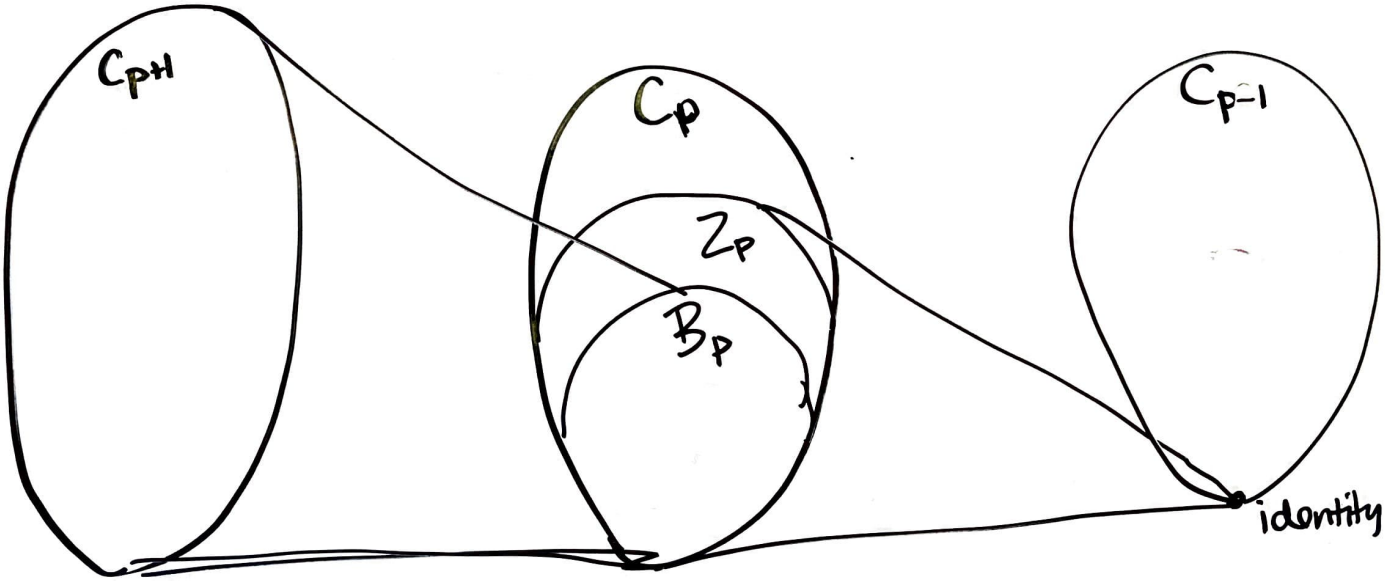
$Z_1 = \ker(\partial_1) = \langle [ab] + [bc] + [ca] \rangle$

$B_0 = \text{Im}(\partial_1) = \{b-a, c-b\rangle$

$Z_0 = \ker(\partial_0) = C_0 = \langle [a], [b], [c] \rangle$

$Z_p = \ker(\partial_p)$ ← the cycle group
" something w/out boundary

$B_p = \text{im}(\partial_{p+1})$ ← the boundary grp
" things that exist as boundaries of other things.



$$B_p \trianglelefteq Z_p$$

$$\Downarrow$$

$$H_p := Z_p / B_p$$ is well-defined,

$(G, +)$

$H_p = Z_p / B_p := \{ x + B_p \mid x \in Z_p \}$ ← a set of cosets

↳ two cycles are the same if they differ by a boundary.

$H_0 = Z_0 / B_0 = \langle a, b, c \rangle / \langle b-a, c-b \rangle \cong G^3 / G^2 = G^{3-2=1}$

$= \langle a \rangle$

↳ What things are equivalent?

$[a] \stackrel{?}{=} [b]$ in H_0

Well, do they differ by a boundary?

Is $[a]-[b] \in B_0$? YES! $\partial([ba]) = a-b$

$[a] = 2[a]$ in H_0?

is $[a]-2[a] = -[a] \in B_0$? No

$[a]-2[b]$ is in $B_0 \Leftarrow [a]-2[a]$ is in B_0 –

toddler

Baby example:

$G_1 = \langle a, b \rangle = \{ g_1 a + g_2 b \mid g_1, g_2 \in G \}$

$G_0 = \langle a \rangle = \{ g a \mid g \in G \}$

$\langle a, b \rangle / \langle a \rangle = \{ (g_1 a + g_2 b) + \langle a \rangle \mid g_1, g_2 \in G \}$

with $\mathbb{Z}$ coef:

$= \{ (\quad g_2 b) + \{ \ldots, (g_1 - 2)a, (g_1 - 1)a, (g_1)a, \ldots \} \mid g_1, g_2 \in$

all coef. are still there, just

$(g_1 a + g_2 b) + \langle a \rangle$

$(g_1' a + g_2' b) + \langle a \rangle$

} these are the same if $g_2 = g_2'$

Another way to think about this:

$x, y \in G_1 \quad x + G_2 = y + G_2 \iff x - y \in G_2$ "Their d

the quotien

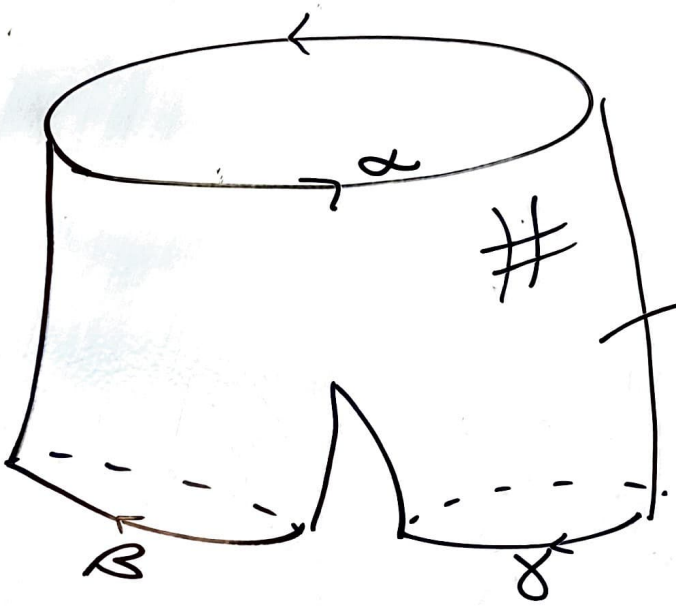
boundary.

$G^3/G^2 = G^{3-2=1}$

$= a - b$

2D example

$K =$



surf S

$\partial_2(S) = \underbrace{\alpha+\beta+\gamma}_{\in B_1}$

Note $\alpha, \beta, \gamma \in Z_1(K)$

in H_1

$\{ \alpha + B_1, \beta + B_1, \gamma + B_1,$

$\ldots, (\alpha+\beta)+B_1, \underbrace{(\alpha+\beta+\gamma)+B_1}_{=b \in B_1 \,=\, B_1}, \ldots$

$\ldots, \underbrace{(0\alpha+0\beta+0\gamma)+B_1}_{=\,B_1}, \ldots \}$

$\alpha + B_1$ and $(\beta-\gamma)+B_1$ are the same in H_1 $\Big\}$ $\alpha-(-\beta-\gamma) \in B_1$